

Erdős Problem #918

EP-918

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Is there a graph with $\aleph_2$ vertices and chromatic number $\aleph_2$ such that every subgraph on $\aleph_1$ vertices has chromatic number $\leq \aleph_0$?

Is there a graph with $\aleph_{\omega+1}$ vertices and chromatic number $\aleph_1$ such that every subgraph on $\aleph_\omega$ vertices has chromatic number $\leq \aleph_0$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>